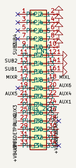
POWER ENTRY, LOGIC SUPPLY, GROUNDS, DECOUPLING

R1 10R · C1 220u 25V · C2 100n · R2 10R · C3 220u 25V · C4 100n · R3 10R · C5 47u 25V · R4 10R · C6 47u 25V · NT1 AGND–LGND star · CN4 Ground spade 6.3mm · R5 10R GND cancel · U1C OPA1612 · C20 100n · C21 100n · U2C OPA1612 · C22 100n · C23 100n · U3C OPA1612 · C24 100n · C25 100n

U5C OPA1612 · C29 100n · C30 100n · U6C OPA1612 · C31 100n · C32 100n · U7C OPA1612 · C33 100n · C34 100n · U8C OPA1612 · C35 100n · C36 100n · U4C LM393 · C26 100n · #FLG01 PWR_FLAG · #FLG02 PWR_FLAG · #FLG03 PWR_FLAG · #FLG04 PWR_FLAG · #FLG05 PWR_FLAG · #FLG06 PWR_FLAG

MOTHERLOOM (Delta pinout; one header, crimped socket on a daisy-chained ribbon)



LINE INPUT — ground-compensated receiver (inverting), Ø switch SW15: latched = phase reversed

J1 Line in TRS · C8 100p C0G · C9 100p C0G · R10 1M · SW15A Phase · SW15B Phase · R6 10K 0.1% · R7 10K 0.1% · C7 33p C0G · R8 10K 0.1% · R9 10K 0.1% · U1A OPA1612

GAIN −15..+15 dB, 0 dB at the detent (inverting; pot pin 3 = CW = input end)

R45 2K2 1% · RV9 Gain 10K lin CD · R46 2K2 1% · C37 22p C0G · U1B OPA1612 · C38 47u NP 25V · R47 47K

HPF unity Sallen-Key Butterworth, −3 dB 87 Hz SW11: latched = HPF in (unlatched shorts the caps)

C39 47n C0G · C40 47n C0G · R48 27K · R49 56K · U5A OPA1612 · SW11A HPF · SW11B HPF

INSERT POSITION JUMPERS 1-2 = insert post-EQ (default); 2-3 on all three = insert pre-EQ

JP6 EQ input · JP5 Insert send source · JP7 PFL source

EQ Delta ED3344A/ED3151 values: HF shelf (U5B), HI-MID (U6A), LO-MID (U6B), LF (U7A); CW = boost

R50 33K · R51 33K · C41 33p C0G · U5B OPA1612 · RV10 HF 10K lin CD · C42 6n7 C0G · R52 5K6 · C43 6n7 C0G · R53 6K8 · R54 22K · R55 22K · U6A OPA1612 · RV11 HI-MID gain 10K lin CD · C44 3n3 C0G · R56 6K8 · RV12 HI-MID freq 100K C dual · R57 6K8 · C45 1n5 C0G · C58 22K · R59 22K · U6B OPA1612 · RV13 LO-MID gain 10K lin CD

C46 15n C0G · R60 6K8 · RV14 LO-MID freq 100K C dual · R61 6K8 · C47 6n8 C0G · R66 22K · R67 22K · U7A OPA1612 · R62 3K3 · R63 3K3 · RV15 LF 10K lin CD · C48 150n film · R64 1K5 · C49 2.2u film · C50 100n C0G · R65 5K1

EQ IN SW12: latched = EQ in (unlatched = bypass from the EQ input); green EQ LED

SW12A EQ · SW12B EQ · R69 4K7 · D5 EQ green · C51 47n NP 25V · R68 47K

INSERT (send = tip, return = ring, ring normalled to send) + return buffer

R11 75R · J2 Insert TRS switched · C10 47u NP 25V · R12 47K · U7B OPA1612 · C11 220u NP 25V · R13 47K

OVERLOAD DETECTOR (-+18 dBu)

R39 1K · D1 1N4148 · C12 1u · R14 470K · R40 10K · R41 10K · U4A LM393 · R15 1K · D2 OL red · U4B LM393

PFL (pre-fade) SW3: A-B = ON

R32 22K · SW3A PFL · SW3B PFL · R33 100K · R34 4K7 · D3 PFL red

ON / SOFT MUTE (2x J112 shunt, ~20 ms ramp) SW4: latched = channel ON, green LED

R16 4K7 · Q1 J112 · R17 4K7 · Q2 J112 · U3B OPA1612 · SW4A ON · R42 220K · C28 100n · SW4B ON · R43 4K7 · D4 ON green

FADER + POST-AMP (+10 dB; omit R38 for unity) JP1 = VCA bypass

RV1 Fader 10K log 100mm · C13 2.2u · JP1 VCA bypass · R18 100K · J3 VCA module socket · J4 VCA control in · R44 100K · U2A OPA1612 · R19 4K7 · C14 100p C0G · R38 2K2 · C15 220u NP 25V · R35 47K

AUX SENDS PRE switch per pair (latched = pre); jumpers 1-2 = pre-fader, 2-3 = pre-EQ. SW13 DIR: aux 6 pot -> direct out

JP2 Aux1/2 pre source · SW8A PRE 1-2 · SW8B PRE 1-2 · JP3 Aux3/4 pre source · SW9A PRE 3-4 · SW9B PRE 3-4 · JP4 Aux5/6 pre source · SW10A PRE 5-6 · SW10B PRE 5-6 · RV2 Aux1 10K log · R20 22K · RV3 Aux2 10K log · R21 22K · RV4 Aux3 10K log · R22 22K · RV5 Aux4 10K log · R23 22K · RV6 Aux5 10K log · R24 22K · RV7 Aux6 10K log · R25 22K · SW13A DIR

SW13B DIR

DIRECT OUT (level = aux 6 pot when DIR is latched; impedance-balanced TRS)

R70 100K · U8A OPA1612 · R71 75R · C52 47u NP 25V · R72 47K · R73 75R · J5 Direct out TRS · U8B OPA1612

PAN + PAN BUFFERS (AC-coupled outputs)

RV8 Pan 10K dual · U2B OPA1612 · U3A OPA1612 · C16 220u NP 25V · R36 47K · C17 220u NP 25V · R37 47K

PAN SWITCH SW14: latched = subgroups panned; unlatched = subgroups get POST, unpanned (MIX always panned)

SW14A PAN · SW14B PAN

ROUTING MATRIX (A-B = assigned; off grounds the resistor) 4K7 summing

R26 4K7 1% · R27 4K7 1% · SW5A Assign L-R · SW5B Assign L-R · R28 4K7 1% · R29 4K7 1% · SW6A Assign 1-2 · SW6B Assign 1-2 · R30 4K7 1% · R31 4K7 1% · SW7A Assign 3-4 · SW7B Assign 3-4

Nets joined by labels: tidy wiring in Eeschema as needed
Generated from channel_strip_schematic_spec.md v4.0

Sheet: /
File: channel_strip.kicad_sch

Title: Channel strip — goa console (Delta Theatre input without the mic stag	
Size: User	Date: 2026-10-19
KiCad E.D.A. 10.0.7	Rev: 4.0
	Id: 1/1